

How the construction of mental models improves learning

Monica Bucciarelli

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Abstract In this paper, I present a framework where possible relations between learning and mental models are explored. In particular, I'll be concerned with non-symbolic gestures accompanying discourse and their role in inducing the construction of models and therefore deep comprehension and learning in the listener. Also, I'll be concerned with cognitive and socio-cognitive conflicts and their roles in inducing construction of alternative models of a problem and therefore in learning to reason. Human ability to learn is of great importance for individuals interested in change. Indeed, to learn both declarative and procedural knowledge means to change, and in order to be able to intervene on change in a desired way it is necessary to have a theory of the mental representations and processes involved in learning and a theory of the communication and contexts that favour learning.

Keywords Mental models · Learning · Gestures · Cognitive conflict

1 Introduction

Learning can be concerned with the acquisition of new knowledge, thus with declarative knowledge, or with being able to improve our own performances, thus with procedural knowledge (Bara 1995).

Declarative knowledge is a theory of the world, conceived of as a set of all the conceptual entities describing, in the form of propositions, classes of objects (e.g., theater), relationships (e.g., people are in the theater), processes

M. Bucciarelli (✉)
Centro di Scienza Cognitiva e Dipartimento di Psicologia,
Università di Torino, Via Po 14, 10124 Torino, Italy
e-mail: monica@psych.unito.it

(e.g., the play), behavioural norms (e.g., one does not smoke in the theater), etc. It does correspond to *knowing that*, and comprises what may be expressed in words about the object or about the state of affairs at issue. Declarative knowledge represents what people believe themselves to know about any given entity in the world, and it is, in principle, conscious; it can be expressed verbally, and it may be reflected upon voluntarily. It is transparent, in the sense that it can be read by the rest of the system's procedures, and if activated intentionally it may become conscious. For example, people familiar with the definition of "mammal" are also aware that they know the definition if they focus their attention on it. All our educational establishments and a large part of professional training are centered on declarative knowledge, even if this has greater weight in the training of, for example, a lawyer than in that of a pianist.

Procedural knowledge refers to the knowledge that a system possesses and that enables it to interact effectively with the world, even though such information is not represented in an explicit fashion so as to be directly readable by other parts of the system. It does correspond to *knowing how* to act. An intentional change requires the existence of an agent executing a clearly defined sequence of operations in order to generate the desired effect. An example of this type of change is giving an invited lecture, where the preconditions consist in there being a conference, the agent is an invited guest, the operations to be executed are those that permit the guest to go to the conference, and the effect is that among the invited speakers at the conference there be the agent himself. This part of procedural knowledge is transparent because the system has free access to such knowledge. A large part of procedural knowledge is, instead, constituted by opaque procedures—methods of acting that are triggered automatically, without requiring control or attention. Typical examples are riding a bicycle or solving a problem—all things that an expert knows well and can do perfectly, but which can only be translated into verbal terms with gross approximations. Opaque procedures operate unconsciously and their reconstruction can take place only a posteriori, in an incomplete and arbitrary manner. Summing up, procedural knowledge corresponds to knowing how to act in a given situation, living the situation from within, and not standing outside of it to speak about it or to reflect on it.

In practice, the distinction between declarative and procedural knowledge is less clear cut than in the definition. And indeed, their relationship is complex and hard to define, corresponding broadly to the relationship between experience and symbol. To infer the explicit knowledge equivalent of a piece of tacit knowledge corresponds to building a theory of a phenomenon, with the proviso that the phenomenon is introspective instead of pertaining to the external world. Just as a propositional theory may be constructed starting from tacit knowledge, so may procedural knowledge be built from explicit knowledge. For example, I can read a book on strategies in reasoning and then do my utmost to apply those principles to the way I solve reasoning problems, trying to keep the procedures thus generated under strict control until they are efficiently stored and coordinated.

In this paper, I will focus on two specific sorts of learning: learning from a discourse (declarative knowledge) and learning to reason (procedural knowledge). In my proposal, deep learning involves the construction and manipulation of models, which are mental processes postulated by mental model theory (MMT; Johnson-Laird 1983, Johnson-Laird and Byrne 1991). MMT claims that individuals construct models when they have to draw inferences. In particular, it is assumed that people construct a model for each sentence, integrate such models, and consider what if anything follows. Eventually, when they behave rationally, people look for a model that falsifies the conclusion initially drawn in order to draw a new necessary conclusion, valid in all the integrated models of the premises. Consider, for example, the sentence “The circle is on the left of the triangle”. An analogical representation could be the following one:

$$\text{O} \quad \Delta$$

where there are tokens for the elements in the sentence, and they are in the same relation as the elements in the sentence. Now consider the description “the star is on the right of the triangle” and the respective model:

$$\Delta \quad *$$

The integration of the two representations through the overlapping of the common elements produces the following integrated mental model:

$$\text{O} \quad \Delta \quad *$$

from which it is possible to extract information that was no explicit in the sentences, and it is possible to infer that “the circle is on the left of the star”, or “the star is on the right of the circle”. As there are no alternative integrated models in which the conclusion is falsified, the conclusion itself is valid. When it is possible to construct alternative integrated models, the conclusion has to be supported by all of them. If no such conclusion exists, then “no valid conclusion” follows from the sentences. The example concerns the domain of relational reasoning, but the mental model theory explains and predicts, on the basis of the same assumptions, the performance of individuals of different ages with different inferential tasks (see, for a review, García-Madruga et al. 2000; Johnson-Laird 2001).

The construction of models is effortful with respect to the encoding of the same information through propositional representations (i.e., simple verbal comprehension), but it recompenses in terms of possibility to draw inferences. As it should be clear by the example above, reasoning means “going beyond” what is explicitly asserted by some premises. This occurs within the constraints posited by necessity in deductive reasoning (i.e., the conclusion must hold in all the models of the premises) and within the constraints posited by plausibility in everyday reasoning (i.e., the conclusion must hold in at least one

model of the premises). In particular, in discourse comprehension, through the construction of models the listener can infer implicit information that is possibly, but not necessarily, true. The listener establishes the relationships and the details that are missing in the set of propositional representations that constitute the discourse, whose explicit content is usually a scheme of facts. Thus, for example, given the sentences “It’s raining” and “Claudia has no umbrella”, we might derive a series of plausible inferences, such as “She buys an umbrella”, or “She decides to stay home”.

I will argue that deep learning involves drawing inferences by means of the construction and manipulation of models; the learner has to give sense to a series of information within a coherent system of meanings and hence construct models from which inferences can be drawn. Biggs (2001) claims that students’ approaches to learning depends upon their perceptions of the context, their own goals, feelings of self-efficacy, and so on. In this sense, it is inappropriate to categorize students as surface or deep learners. However, it is possible to classify in these terms each specific approach to learning. Thus, Biggs proposes that some approaches involve low cognitive level activities (e.g., memorize, identify), some approaches involve intermediate level activities (e.g., comprehend main ideas, relate), and others high level activities (e.g., apply to far domains, reflect). Within the theoretical framework, I propose that high-level activities consist of the construction and manipulation of mental models.

2 Learning within a mental model framework

Within the mental model research paradigm, learning declarative knowledge means to construct mental representations of increasing complexity (Halford 1993, Johnson-Laird 1983). The possibility to draw inferences, which is offered by the construction of mental models, denotes the full comprehension of a state of affairs. Deep comprehension and learning always require the construction of models. Given this framework, what favours the construction of models in the learner? I will argue that non-symbolic gestures accompanying discourse have a role in the construction of a model of the declarative knowledge it conveys. Also, I will argue that cognitive conflicts have a role in learning to reason, a sort of procedural knowledge, as they induce the construction of alternative models. Thus, both non-symbolic gestures and cognitive conflicts contribute to a meaning’s construction by leading us to enrich our mental representations of a discourse or a problem.

Many studies in the literature are consistent with the assumption that non-symbolic gestures affect meaning construction in communicative interactions. Some studies investigate how a partner in the communicative interaction attributes communicative intentions to an actor gesturing while proffering no words. In particular, children (see Bucciarelli et al. 2003), as well as adults (Bara et al. 2000), have the pragmatic competence to attribute to an actor the intention to be sincere, to deceive, and to be ironic when communicating

through gestures. Other studies investigate how a combination of words and gestures works in meaning construction. Goldin-Meadow and Butcher (2003), for example, find that in the one-word period children produce gesture-speech combinations in which gesture conveys information that is different from the information conveyed in speech. Combinations of this sort are, clearly, the products of a unified gesture-speech system, in that they allow children to express two elements of a sentence, one in gesture and one in speech. In mental models' terms, non-symbolic gestures are a different modality from speech, but both of them coordinate in meaning construction, through the possibility they offer to construct and enrich our mental models (Bucciarelli 2005). The end-product of this process is a holistic representation of the discourse, where traces of the modality in which particular information was introduced is lost (see Glenberg and Langston 1992, Rickheit and Wrobel 1999).

Also, many studies in the literature are consistent with the assumption that cognitive conflicts affect our way to comprehend the deep structure of a problem, as they induce us to think about a problem from multiple perspectives. Consider, for example the selection task devised by Wason (1966). In the original version of the task, the participants have in front of them four cards:

A B 2 3

They know that each card has a letter on one side and a number on the other side. Given a conditional rule: *If a card has the letter "A" on one side then it has the number "2" on the other side*, their task is to select those cards that have to be turned over to discover whether the rule is true or false about the four cards. Most people select the A card alone, or the A and two cards. They fail to select the 3 card; yet, if it has an A on its other side, the rule is false. When the conditional lays down a deontic obligation, and the task is to choose potential violations of the obligation, both adults and children make a greater number of correct selections (e.g., Johnson-Laird et al. 1972; Kroger et al. 1993). For example, given the deontic conditional from Griggs and Cox (1982): *If a person is drinking beer then that person must be over 19 years of age*, most adults select the correct cards representing potential violations, i.e., the cards representing *a person drinking beer* and *a person less than 19 years*.

In the so-called "reduced array" selection task, the participants make their selections only from two cards: given a rule of the form, *If A then B*, they select only from the two cards corresponding to the B and the not-B cases (Wason 1968). A striking discovery was that even 9- to 10-year-old children can get this version of the selection task right if it is framed deontically (see Girotto et al. 1988). Cummins (1996) replicated these results but with 3- to 4-year olds.

As Bucciarelli and Johnson-Laird (2001) point out, deontic versions, as well as other versions sorting facilitatory effect (see Cheng and Holyoak 1989; Manktelow and Over 1990, 1991; Johnson-Laird and Byrne 2002), make

salient the other's mental states such as beliefs and expectations that might be in conflict with our own. Thus, they offer the possibility to have a multiple perspective on the problem that would result in a deeper understanding of the structure of the problem.

The paper deals with learning of declarative and procedural knowledge in two parts (Sect. 3, 4). Section 3 is concerned with learning from discourse; Sect. 4 is concerned with learning to reason.

3 Learning from discourse

3.1 Figures, diagrams, and learning

Within cognitive psychology, the relation between construction of a model of the discourse and learning has not been much investigated. On the contrary, many attempts were made to characterize the mental representation constructed during *text* comprehension. A general assumption is that it requires the construction of a referential representation of the meaning of the text: a situation model (see Van Dijk and Kintsch 1983) or a mental model (see Johnson-Laird 1983). In recent years the terms mental model and situation model have often been used synonymously as they serve similar functions. I will prefer the term mental model as it comes with a series of assumptions on the mental representations and processes underlying the inferential ability.

Also, it has been claimed that learning might benefit from visual presentation of material. For example, research conducted during the 1970s and 1980s supported the assertion that carefully constructed *text illustrations* generally enhance learners' performance on a variety of text-dependent cognitive outcomes. Most recent researches still support this claim (see, e.g., Carney and Levin 2002; Cuevas et al. 2002; Moreno and Mayer 2002; Stenning 2000; Vekiri 2002).

What makes figures and diagrams accompanying text relevant to learning? As Antinucci et al. (1994) claim, a communication that reproduces the structure of the domain of knowledge object of learning should improve learning. They argue that an analysis of the sort of communication used in didactic contexts cannot disregard the structure of the knowledge field to be learned. If we define the problem in terms of the relationship between knowledge field structure ("KS" for "knowledge structure") and communication structure ("CS" for "communication structure"), then it follows that CS will be more appropriate as its relation with KS is simple, transparent, and immediate; thus the CS will be more appropriate as it will be isomorphic to the KS that it must transmit.

Antinucci et al. argue that the isomorphism between CS and KS is the key for the effectiveness of both the communicative and learning processes, in that it simplifies the fundamental difficulties of these processes: the translation from CS to KS. As Stenning (2000) sets the hypothesis: the fundamental

distinction between direct and indirect systems of representation is that indirect systems have an abstract syntax interposed between representation and represented entities (see also McKendree et al. 2002). As Antinucci et al. argue, if we could structure communication in a way identical to the way in which the knowledge field that it has to transmit is structured, then the organization of the knowledge field would be apparent just from “watching” CS. The problem becomes how to construct communication structures that are as isomorphic as possible to the structures of the knowledge fields they transmit. From the point of view of Antinucci et al., figures and diagrams accompanying text are relevant to learning because they reproduce the structure of the domain of knowledge object of learning.

3.2 Non-symbolic gestures and learning

Also, there is a series of experimental evidence that induces us to think that non-symbolic gestures are strictly related to understanding and learning.

The experimental literature suggests that gesture has a role for both the observer and the learner. As regards the role for the observer, Goldin-Meadow (2001) claims that gesture, particularly when considered in relation to the speech it accompanies, can be used to identify when a child is in a transitional state. Thus, gesture could be a general mechanism of change (see also Schwartz and Black 1996). In line with this assumption, Siegler (1976) found that although both 5- and 8-year olds used a weight-only rule in solving a series of balance-scale problems, the 8-year-olds produced head movements that indicated they were also aware of the weights’ distance from the fulcrum; the 5-year olds gave no such evidence and made significantly less progress on the task. Also, Alibali and Goldin-Meadow (1993) observed children while studying mathematical equivalences and they distinguished those gesturing while studying from those not-gesturing. After a 2-week period, children who gestured during the process showed a tendency to outperform children who did not gesture. Further, in a conservation study, Church (1999) found that it is not the different number of strategies that matters in predicting learning, but whether those strategies are produced in gesture. The very same experimental literature just mentioned suggests that, in addition to signaling the learner’s cognitive state to others, gesture may have a role in the learner; in particular, it might function in some beneficial way for the learner. Further evidence comes from a study by Alibali et al. (2001). They hypothesize that gesture highlights perceptual-motor information in the conceptual planning of speaking. One implication of this view is that when gesture is prohibited, people should be less likely to use perceptual-motor thinking. The results of their study confirm this expectation. The authors conclude that gesture is involved in the conceptual planning of speech, and producing gesture may help speakers decide what to say (see also Alibali et al. 2000).

But, why might gesture be associated with learning? Goldin-Meadow (2001) claims that:

- (a) *Gesture may ease the cognitive burden.* In particular, gesture can reduce cognitive effort perhaps in the same way that writing a problem down can reduce the effort needed to solve the problem. The act of conveying information, whatever that information might be, in a second modality may make the task easier for the child. Using two modalities rather than one may add redundancy—when gesture conveys the same information as speech—or it could make the task easier by providing a vehicle that allows the child to express thoughts that she/he cannot yet express in speech.
- (b) *Gesture is where children experiment.* Either gesture is an ideal place to try out untried ideas in that there are no social constrictions on the gestures people produce or is the ideal place for experimentation because the ideas themselves are easier to express in the manual modality; this would be particularly true for domains such as mathematics, which lend themselves to visual thinking (see, e.g., Hadamard 1945).

In this paper, I am concerned with learning rather than with development. However, I believe that the developmental literature on gestures is relevant to the topic. Indeed, I do not make any in-principle distinctions between the mechanisms underlying transitions in children's cognition and many other cognitive changes occurring later in life, such as in the case of learning (see also Simon and Halford 1995). As I assume that similar mechanisms may characterize changes occurring during development and learning, in the next section I will reinterpret the claims by Goldin-Meadow within my proposal for learning by mental models.

3.3 Mental model perspective

Within my proposal, the isomorphism of figures and diagrams with respect to the knowledge fields they refer to, favour learning of declarative knowledge because they favour the construction of models for such fields. In my view, figures are constituted by analogic elements that favour the construction of models. Consistent with this view, Schnotz and Bannert (1999) also view text and picture comprehension as a task-oriented process of constructing mental representations by mental model construction and model inspection. Also Gyselinck (1996) claims that the mental model theory can provide an account for the beneficial effect of illustrations in text comprehension. Through a review of the literature he shows that pictures facilitate the construction of a mental model, even in domains of discourse that are not exclusively spatial.

Within my proposal I add that gestures accompanying discourse, as well as figures and diagrams, have analogical characteristics that *reflect* the structure of the mental models held by the individual who gestures. As a main consequence, I hypothesize that non-symbolic gestures accompanying the discourse of a speaker favour the listener's model construction. Thus, within my proposal, the assumptions by Goldin-Meadow (2001) on the reason why gestures might be associated with learning can be specified as follows: gestures ease the

cognitive burden, and they allow children to experiment and improve learning because they favour the construction of models. A main function of mental models is the organization of information into a meaningful whole and thus derive the possibility to more easily keep in mind a series of information. As a consequence they relieve an individual's cognitive burden—see point (a) above—and allow the drawing of inferences concerning the information implicit in the discourse, thus improving learning—see point (b) above.

3.4 A litmus test: a study on gestures and learning from a discourse

If my assumptions are correct, we might expect to find out that when non-symbolic gestures accompany discourse, as compared with situations in which no gestures accompany discourse, the listener constructs more elaborate mental models of the discourse. Two main consequences for gesture condition, as we evince from Johnson-Laird 1983, p. 162), are:

- (1) *The listener should retain more information.* Models are easier to remember than propositions, perhaps because they are more structured and elaborated (cf. Craik and Tulving 1975), and require a greater amount of processing to construct (cf. Johnson-Laird and Bethell-Fax 1978).
- (2) *The listener should be more likely to believe to remember information that actually corresponds to inferences made from the information explicit in the discourse.*

Models encode little or nothing of the linguistic form of the sentences on which they are based, and individuals accordingly confuse inferable descriptions with the originals. Propositional representations are relatively hard to remember, but they do encode the linguistic form of sentences. Hence, when they are remembered, the individuals are likely to make a better than chance recognition of verbatim content.

Ilaria Cutica and I devised an experiment to test these predictions (Cutica and Bucciarelli 2003). Thirty-eight adult participants (mean age: 23 years) had to watch the videotape of a professional actor making a discourse. They were randomly assigned to one of two experimental conditions. In Gestures condition, the actor in the videotape accompanies the discourse with non-symbolic gestures, whereas in No-Gestures condition, the actor in the videotape makes the very same discourse without gesticulating. Then, in both conditions participants had to recall as much information of the discourse as they could.

We analyzed the means of correct recollections and the means of recollections that correspond to plausible inferences (Table 1). As a general result, the correct recollections were more than the plausible recollections in both conditions. However, the results of the experiment fully confirm the expectations. Participants in Gesture condition recollected significantly more information with respect to participants in No-Gesture condition (t -test for independent samples: $t(36) = 5.60$, tied $p < .0001$). Also, they recollected significantly more information that corresponds to plausible inferences with

Table 1 Means of sorts of correct recollections and recollections that correspond to plausible inferences in the experiment by Cutica and Bucciarelli (2003)

	Correct recollections	Plausible inferences
Gestures	21.89	1.05
No gestures	15.32	0.21

respect to participants in No-Gesture condition (t -test for independent samples: $t(36) = 2.94$, tied $p = .004$).

More recently, we have replicated the results above, and we have also experimentally validated the prediction complementary to the prediction in the experiment above: a discourse proffered without gesturing, as compared with a discourse accompanied by gestures, results in a better performance of the listener in recognizing the verbatim of the sentences in the discourse (Cutica and Bucciarelli, Submitted: Experiments 2 and 3). To sum up, the results of our experiments suggest that non-symbolic gestures accompanying discourse favour learning of the information in the discourse by favouring the construction of models in the listener.

3.5 Discussion

A case example of learning of declarative knowledge is learning from discourse. I assumed that learning of declarative knowledge involves the construction of models, and I collected evidence in favour of this assumption. In particular, some of this evidence is concerned with the role of gestures accompanying discourse in favouring the construction of models and therefore learning. A criticism that could be raised (Philip Johnson-Laird and Dan Sperber personal communications) is that gestures focus the listener's attention on the verbal expressions they accompany, rather than being vehicles of meanings. An experiment impossible to realize in natural condition would be to see whether the same learning effects can be found when each gesture spontaneously produced in correspondence with a verbal statement occurs in correspondence with a different verbal statement. The impossibility to realize such an experiment rests on the impossibility to ask an actor to gesticulate "changing the co-occurrence of a specific gesture with a specific verbal act".

However, some experimental results in the literature constitute an indirect evidence of the fact that gestures convey information. Garber, Alibali, and Goldin-Meadow (1998), for example, investigate whether children's gestures convey knowledge that the children themselves can recognize in another context. The starting point for this study is the observation that children frequently gesture when they explain what they know, and their gestures sometimes convey different information than their speech does. A nice exemplification is provided by Garber and Goldin-Meadow (2002, p. 818):

consider a child shown two tall thin glasses containing the same amount of water. The water in one glass is poured into a short wide dish and the

child is asked whether the dish has the same amount of water as the full glass. The child says “no” and justifies his incorrect belief by saying “it’s different because this one’s tall and that one’s short” while at the same time gesturing TALL (a flat palm held at the water level of the full glass) and then SHORT (a flat palm held at the water level of the dish). This child is focusing on the height of the water and has indicated his focus in gesture as well as speech. There are times, however, when speakers use their hands to convey information that is *not* found in their speech. Consider another child who gives the same response to the water conservation question in speech – “it’s different because this one’s tall and this one’s short” – but in gesture produces SKINNY (two flat palms held at the sides of the glass) and then WIDE (two flat palms held at the sides of the dish). This child highlights height in her speech but width in her gesture – she has produced a gesture-speech mismatch.

Now, Garber et al. (1998) ask to forth-grade children to explain their solutions to a set of math problems. Then, the authors identified the solution procedures that each child conveyed only in gesture (and not in speech) during the explanations, and examined whether those procedures could be accessed by the same child on a rating task that did not involve gesture at all. Children rated solutions derived from procedures they conveyed uniquely in gesture higher than solutions derived from procedures they did not convey at all. Thus, gesture can be considered a vehicle through which children express their knowledge.

Also, Goldin-Meadow and Sandhofer (1999) explore whether gesture affects how accurately its accompanying speech is interpreted. In particular, they asked adults to observe children while participating in a series of Piagetian conservation tasks. The authors gave adults a checklist for each task the child performed in order for them to be able to assess their knowledge online. The list contained all of the explanations children typically give on tasks of this sort. The adults’ job was to check off as many explanations as they thought the child had conveyed. They found that the adults checked explanations that the children conveyed *only* in their gesture from 32 to 44% of the time. These results suggest that the adults were able to read the children’s gestures. Also, Goldin-Meadow and Sandhofer found that the adults correctly checked off explanations conveyed in speech significantly more often when they were accompanied by a matching gesture (i.e., gesture conveyed the same explanation as the speech it accompanied) than by a mismatching gesture (i.e., gesture conveyed a different explanation). Differently from Cutica and Bucciarelli (2003), Submitted, and Goldin-Meadow and Sandhofer (1999) did not find evidence that gesture improves a listener’s ability to recognize a message produced in speech if gesture conveys the same message. However, they found evidence that gesture diminishes a listener’s ability to recognize a spoken message if gesture conveys a different message (see also Kelly and Church 1998).

4 Learning to reason

4.1 Cognitive conflict and cognitive changes

Bruner et al. (1966) introduce the term “conflict” in giving an account of the development of the ability to reason in children. In particular, they focus on the interaction of the individual with objects. The main mechanism of development is the conflict between different modalities of representation; when two systems of representations do not correspond, like in Fig. 1, when in other words what one sees does not correspond to what one says (e.g., the iconic representation of the different levels reached by the water does not correspond to the symbolic representation of the identity of the water as expressed verbally), the child executes clear-cut revisions to his way of solving problems.

Other theorists, still within a developmental perspective, underline the importance of social interactions in cognitive development and claim that the first stages of thinking development are deeply social, as individual abilities become structured thanks to interaction with others (see, e.g., Vygotsky 1978). Doise and Mugny (1984) claim that the cognitive progress is triggered by perturbation in the interactions with other individuals, for which they introduce the term “socio-cognitive conflict”; they empirically found that the experience of the socio-cognitive conflict assists children in reaching the correct solutions to certain problems. For instance, reasoning with a conservation task otherwise hard for a child, can be facilitated if the situation is connoted socially. Doise and Mugny (1984) connoted socially the situation by saying that two glasses of juice were a premium for two children who were equally nice, thus introducing the social norm known by the child according to which the two children have the right for the same premium given the same behaviour. The results reveal that children were able to conserve the substance also when subsequently tested with two different glasses. Thus, the social connotation of the cognitive task can produce cognitive progress also when the individual works alone. Also, reasoning with a number conservation task can be facilitated by asking the child to explain the point of view of the individual who gives the correct answer (Siegler 1995). Thus, when working alone, children at a certain stage of development are unable to perform cer-

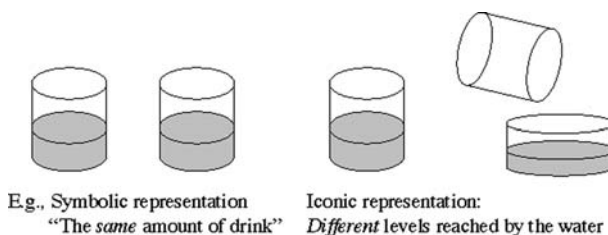


Fig. 1 An example of conflict between different modalities of representation

tain tasks because they cannot decenter, i.e., they focus on certain aspects of problems excluding others, more important aspects; however, the same children exposed to differing points of view with regard to such problems find it easier to decenter. It follows that social interactions would provide an obvious forum for the possibility of exchange of perspectives. The results of the experiment by Doise and Mugny are twofold. First, when individuals are in the initial phase of the acquisition of a new notion, working in groups is more effective than working individually in producing real cognitive progresses, in that stable in the time and generalized to other tasks. A noteworthy progress occurs, for example, in the acquisition of the notion of conservation in groups of 6-year olds. Second, both the interactions with participants at the same cognitive level as well as with participants at different cognitive levels are useful. This means that it is not necessary to compare ourselves with correct responses in order to have cognitive progress, rather, what is important is to compare our own point of view with another one which is different (more or less correct) in order to feel the need of cognitive restructuring (but cf. Roazzi and Bryant 1998).

If conflicts play a role in cognitive restructuring, it remains to be explained which are the mechanisms at the basis of cognitive change.

4.2 Mechanisms of cognitive change

Karmiloff-Smith (1992) proposes an account of cognitive change in terms of change in the sort of mental representations. She argues that, with increasing age, children's representations become progressively more manipulable and flexible. The reason why this change is possible relies on the fact that initially the implicit information *in* the mind would be in the form of procedures (and therefore implicit), and through a process referred to as representational re-description (RR) such information becomes explicit knowledge *to* the mind (and therefore information manipulable and flexible).

Karmiloff-Smith argues that cognitive changes can be characterized through two main dimensions: exogenous/endogenous and implicit/explicit (see Table 2).

Within the first phases of the development, but also within the first phases of a learning process, cognitive changes are triggered by exogenous causes. The child focuses predominantly on information from the external environment, and representations are in the form of procedures for analyzing and responding to stimuli in the external environment. At this level the infor-

Table 2 Cognitive change

Phases of development/learning	Triggers	Form of representations
Initial	Exogenous	Procedures, implicit
Subsequent	Exogenous, endogenous	Declarative knowledge, explicit

Triggers and forms of representation in the initial and subsequent phases of development/learning according to Karmiloff-Smith (1992)

mation embedded in procedures remain implicit. This gives rise to the ability to compute specific inputs in preferential ways and to respond rapidly and effectively to the environment.

Once re-description has taken place and explicit representations become manipulable, the child can then introduce violations to her data-driven, veridical descriptions of the world. Violations which allow for pretend play, false belief, and the use of counterfactuals. Cognitive changes can now be provoked by endogenous causes.

Consistent with the claim that much learning results in the possibility to have explicit access to our way to represent information, is the claim by McClelland (1995) according to which much knowledge that developmental psychologists study may not be propositional, it may be stored in the form of connections, and it may be implicit, namely opaque, to verbal description.

My proposal, as well as those by Karmiloff-Smith and McClelland, is concerned with learning in naïve individuals who do not have explicit methods to approach a specific situation, rather than with skill acquisition, where rules have to become familiar to an individual. In case of skill acquisition, as proposed by Anderson in his computer model ACT (Anderson 1983), the end-phase of the learning process is implicit, procedural knowledge, rather than declarative knowledge. In ACT, the initial phase of the learning process is characterized by the fact that individuals think about the explicit rules for the implementation of the procedure. Thus, in skill acquisition, where external methods can be provided to accomplish a task, learning proceeds from declarative to procedural knowledge, like in the case of reading a book of logic for learning to reason.

The more general assumptions underlying my proposal is that the ability to construct and manipulate models are procedures built into in our cognitive architecture, and they play an extremely important role in the initial phases of the learning process. The ability to intervene on the learning process by applying methods or strategies is a meta-cognitive ability that develops subsequently. Such a meta-cognitive knowledge is declarative, and it can be utilized to monitor our way to function procedurally. The meta-cognitive control is one of the main dimensions of the cognitive change occurring during the development (Kuhn 1989, 2000; Kuhn et al. 2004) as well as during the process that leads to becoming an expert in a particular field (Byrne and Handley 1997; Johnson-Laird et al. 2000).

4.3 Mental model perspective

Within the mental model research paradigm, the construction and manipulation of models respond to the individual's need to maintain the cognitive system free of contradictions. I assume that to be aware of contradictions is a first step in the process of conceptual change (see also Limón and Carretero 1997). Thus, models would allow to maintain the cognitive system free from contradictions or, in my proposal, from “cognitive conflicts”. I want to argue that the cognitive conflict favors the construction of alternative models of a

perceived or described situation, thus favoring learning to reason. The construction of alternative models can be favored by interaction with objects (see Bruner et al. 1966), as well as by real or imagined interactions with intentional agents (see Doise and Mugny 1984). In both cases, in the initial phase of learning to reason the construction of alternative models would be triggered at a procedural level by some exogenous cause or some external obstacle to our way of acting in the world (see also Karmiloff-Smith 1992). As Popper (1982) stated, the obstacle is the missed fulfillment of an expectation, and its importance relies on the fact that it produces a disappointment and it forces us to change our mind. Kelly (1955) defines an obstacle an invalidation; in my view, it forces us to construct alternative models.

In line with Karmiloff-Smith, we can distinguish two phases in the process of learning as it applies to reasoning. In the *initial phase*, the construction of alternative models is induced by cognitive conflicts and socio-cognitive conflicts triggered by *exogenous* causes. In *subsequent phases* conflicts may be triggered by *endogenous* causes. This would be possible because with development, knowledge becomes more accessible. The mind becomes able to exploit internally the information already stored, by re-describing its representations or, more precisely, by iteratively re-representing in different representational formats what its internal representations represent. For this reason, through a RR process, individuals come to explicitly conceive the utility of constructing alternative models, and this awareness would constitute an endogenous cause of conflicts. Thus, in adults or experts the construction of alternative models may be triggered either by external factors or by internal factors. Let us consider how these assumptions apply to reasoning when testing hypothesis in the Wason's selection task introduced above.

In my view, data in the literature on the selection task suggests that deontic contexts can be considered *exogenous causes* that trigger socio-cognitive conflicts, as when dealing with the deontic version of the task individuals of any age tend to envisage the counter-examples to the rule, i.e., they tend to construct alternative models of the problem. Further, data in the literature suggest that abstract contexts do not sort facilitative effects. And indeed, individuals differ in their ability to deal with the abstract version of the task, and their performance correlates with IQ measures, contrary to their performance with the deontic version (Stanovich and West 1998). We might consider the IQ an *endogenous cause* that triggers conflicts in dealing with the abstract version.

It is plausible that, as suggested by Bucciarelli and Johnson-Laird (2001), deontic contexts trigger innate procedural knowledge concerning the theory of mind, i.e., the ability to attribute mental states, such as beliefs and desires, to oneself and to others, and to recognize that the mental states of others do not necessarily correspond to one's own (Premack and Woodruff 1978, Leslie 1987, Baron-Cohen 1994). Thus, the necessity to take into account possible false beliefs of the others might help one to envisage the false instances of a rule or, in a more general sense, alternative models of a problem.

How could it be possible to induce the construction of alternative models and therefore learning? How could we devise optimal learning contexts within which the experience of conflicts (both cognitive and socio-cognitive) is induced in the learner? Obviously, in such contexts the conflicts would have exogenous causes. I hypothesize that a learning experience that reflects our natural way to cognitively restructure will induce learning. Individuals can first experience conflicts at a procedural level. In particular, the socio-cognitive conflict differs from the cognitive conflict in that it triggers procedural knowledge concerning theory of mind. Thus, we might expect that the best way to induce learning is to provide conflict first at a socio-cognitive level and then at the cognitive level.

4.4 A litmus test: a study on (socio-)cognitive conflict and learning to reason

Katiuscia Sacco and I carried out a study where we compared adults' ability to reason deductively in two different conditions in two experimental sessions, one week one from the other (Sacco and Bucciarelli 2007: Experiment 2). The participants in the experiment, twenty adults (mean age: 24.5 years), deal with syllogisms; for each syllogism, once the participant has stated her own conclusion the experimenter presents an alternative conclusion (the correct one if the participant's conclusion is incorrect, an incorrect one if the participant's conclusion is correct). The participant is then asked to choose the better of the two conclusions. Each participant is tested on two protocols: in the cognitive-conflict protocol (CC) the experimenter states that each alternative conclusion is drawn by lot; in the socio-cognitive-conflict protocol (SCC) the experimenter states that each alternative conclusion has been reached by another participant. Note that, in the SCC protocol, participants are told that the conclusion has been given by *another participant* in order to underline that there was no difference in the social status/authority between the participant and the other person who gave the alternative answer (this procedure aims at avoiding the effects of authority). The core assumption underlying the experiment is that cognitive-conflict and socio-cognitive-conflict induce reasoners to revise their way of solving a problem, thus favouring an improvement in their reasoning performance.

MMT distinguishes between one-model syllogisms and multiple-model syllogisms; the former require the construction of a single model to be solved. On the contrary, when dealing with multiple syllogisms, once reasoners reach a first putative conclusion it is extremely important that they construct all the alternative models of the premises, which together support the logical conclusion. Thus, the construction of alternative models is crucial especially for multiple model syllogisms.

Table 3 illustrates the results for multiple model syllogisms. Participants performed better in the second experimental session than in the first one. This result suggests that learning occurred, and its effects become stable and detectable after 1 week.

Table 3 Percentage of correct performances with multiple model syllogisms in cognitive conflict (CC) and socio-cognitive conflict (SCC) protocols

Multiple-model syllogisms	Cognitive-conflict protocol		Socio-cognitive-conflict protocol	
	First conclusion	Second conclusion	First conclusion	Second conclusion
CC first and SCC second	41	51	45	53
SCC first and CC second	60	67.5	53	68

When participants dealt with them both in first and second session: from Sacco and Bucciarelli 2007: Experiment 2

A related finding is the improvement in performance after the presentation of an alternative conclusion in all sessions, with one exception. In particular, those participants who dealt with the SCC protocol in the first experimental session and with the CC protocol in the second experimental session improve their own performance from first to second conclusion in both sessions. However, those participants who dealt with the CC protocol in the first experimental session and with the SCC protocol in the second experimental session improve their performance from first to second conclusion in the first session, but not in the second session, i.e., when dealing with the SCC protocol. Evidently, the optimal learning process (to learn to construct alternative models) is to start with the alternative conclusion in a SCC context, then move to the alternative conclusion in the CC context.

In particular, when the conflict is first socio-cognitive, and then cognitive, the construction of alternative models enhance learning to reason. The results are in line with my expectations. Thus, maybe the socio-cognitive conflict encountered in the first session motivates a cognitive restructuring that results in an increased ability to construct alternative models in the second, cognitive conflict session.

4.5 Discussion

A case example of learning of procedural knowledge is learning to reason. I have assumed that learning of procedural knowledge involves an increasing ability to construct and manipulate models. Within my proposal, the construction of alternative models is possible thanks to the experience of a (socio-)cognitive conflict. I made a brief review of the literature in support of the claim that conflicts favour an improvement of performance with Piagetian conservation tasks as well as with hypothesis testing tasks.

Also, I have assumed that socio-cognitive conflicts trigger procedural knowledge concerning the theory of mind. This assumption, along with those by Karmiloff-Smith (1992) on the way in which cognitive restructuring occurs, leads to expect that the best way to favour learning is to induce first the experience of socio-cognitive conflict, and then the experience of cognitive conflict. The importance of experiencing meaningful conflicts has been poin-

ted out by Chan (2001). The results by Sacco and Bucciarelli suggest that an especially meaningful conflict has to be social. In the literature, there are few studies on the effect of conflicts in adults, and they are mainly concerned with hypothesis testing tasks (see, e.g., Legrenzi et al. 1991). Our results are more general in that they reveal an effect of cognitive conflict as well, in reasoning with a deductive task.

The efficacy of cognitive conflict depends on various factors we have not analyzed. And these factors can account, at least in part, for the results in the developmental literature, which are inconsistent. Some studies indicate effectiveness of socio-cognitive conflict to bring about cognitive growth (Azmitia 1988; Light and Glachan 1985), whereas in others, children have not improved (Perret-Clermont 1980; Russell 1982). According to Druyan (2001), the reason for the inconsistent results is that the effectiveness of cognitive conflicts depends on both the demands of the task and the developmental stage of the individual. In our study, adult individuals benefited from the socio-cognitive conflict when dealing with complex tasks such as syllogisms. Also, some studies suggest that other factors that can account for inconsistent results in the literature are workgroup size when the conflict is experienced during collaborative work (Fuchs et al. 2000), as well as values and attitudes, learning strategies, cognitive engagement, and reasoning abilities necessary to solve the conflict (Limón 2001). These factors were not considered in the study by Sacco and Bucciarelli (2007), but it is quite obvious that, for instance, in absence of reasoning abilities necessary to solve the conflict neither the cognitive nor the socio-cognitive conflict can sort their effects.

As a general consideration, our results are in line with the literature according to which cognitive conflicts lead us to revise our way to conceive of a problem. In improving our comprehension of a problem, frequently it is necessary to restructure our existing knowledge, a process that is usually referred to as conceptual change (see, e.g., Caravita and Halldén 1994; Vosniadou 1994). Montgomery (1984) argues that cognitive restructuring can be viewed as transforming a mental model into a structural balance that avoids generating cognitive conflicts.

5 Conclusions

Does the construction of mental models improve learning? I have attempted to evaluate this assumption for learning from discourse (declarative knowledge) and for learning to reason (procedural knowledge).

In this paper, I have argued that non-symbolic gestures accompanying discourse induce in the listener the construction of models. And indeed, the investigation by Cutica and Bucciarelli (2003), (Submitted) reveal that when gestures accompany discourse the listener retains more information with respect to a situation in which no gestures are performed. Some studies in the literature lead to the same conclusion. Goldin-Meadow and Sandhofer (1999), for example, found that explanations are better understood when the

accompanying gestures match the explanation than when they mismatch the explanation. However, Krauss, et al. (1991) found that gestures do not improve comprehension of a message. In particular, the requirement of their experimental task was to recognize the words that originally accompanied a series of gestures, and the results of their study reveal that performance was better than chance, but markedly inferior to performance when words were used as stimuli. These data, that seem contradictory, can be explained if we assume that gestures favour the construction of models, whose side effect is the production of plausible inferences. This side effect might penalize the performance depending of the requirement of the task (i.e., memories for the verbatim versus memories for the meaning). Consistent with this explanation, Cutica and Bucciarelli (Submitted: Experiments 2 and 3) find that the listener of a discourse accompanied by gestures, as compared with the listener of a discourse not accompanied by gestures, is favoured in the task to recollect as much information as possible, but is penalized in the task to recognize the verbatim of the discourse. Thus, the answer to the original question could be that gestures facilitate deep learning through the construction of models and, depending on the requirements of the task, they may either favor or impair individual's performance.

Gestures accompanying discourse are visual information accompanying the verbal information. In the literature on multimedia learning environments, the fact that the physically close presentation of both visual and verbal materials induce better learning in students, is accounted for in terms of redundancy. Visual material would add redundancy and therefore would result in better comprehension (see, e.g., Mayer 1997; Mayer and Moreno 2002; Moreno and Mayer 1999). In particular, Mayer and Moreno (2002) claims that the possibility of dual-coding of visual and verbal material in multi-media results in greater learning. However, the experimental evidence is in favour of the fact that gesture do not provide redundancy, rather they provide information not conveyed by words.

Does the construction of alternative models improve learning of procedural knowledge? It is plausible to assume that in the initial phase of development/learning, the construction of alternative models is mostly favoured by socio-cognitive conflicts (see also Doise and Mugny 1984) in that they trigger our procedural knowledge concerning the theory of mind. In the subsequent phases, individuals might have different sensibilities to cognitive-conflict and socio-cognitive conflict. Learning seems to be mainly affected by the encounter of alternative models when they are produced by other intentional agents (see Sacco and Bucciarelli 2007). As a consequence, the improvement in the ability to reason would depend on the source of alternative models (cognitive conflict or socio-cognitive conflict). More in general, the effects of both cognitive conflict and socio-cognitive conflict could depend on the level of cognitive development (see also Vandenplas-Holper 1996) and, in its turn, the fact that conflicts facilitate the construction of alternative models surely depends on the cognitive level.

Gestures and cognitive conflict have been investigated together by Clement and Steinberg's (2002). The authors describe an approach to teaching complex models in science that takes a model construction cycle of generation, evaluation, and modification as an organizing framework for thinking about when to use each of the previous strategies. The results of this study suggest that (a) *discrepant* events produced reactions of surprise and were eventually followed by model revisions, leading us to hypothesize a motivating and guiding role for these events; (b) the subject was able to map and apply an air pressure analogy used for electric potential and continued to exhibit traces of it through the post-test interview. Also, (c) the subject's spontaneous use of similar *depictive hand motions* during the instruction and during the post-test provides initial evidence that the instruction fostered development of dynamic mental models. Thus, the core of the subject's new knowledge are explanatory models at an intermediate level of generality that allow her to run imagistic simulations and to hypothesize a "transfer of runnability" from the analog conception to the model in this case. They conclude on the relevance of using multiple small discrepant events and analogies built into the lessons.

Spiro and Jehng (1990) have presented cognitive flexibility theory within which the ability to construct and switch between multiple perspectives of a domain is fundamental to successful learning. What seems to be shared by non-symbolic gestures accompanying discourse and by cognitive and socio-cognitive conflicts is the fact that they provide multiple perspectives on a single task, i.e., discourse comprehension and problems solution, respectively. Also, gestures, as well as cognitive conflicts, have an effect on the search for coherence. The former concur to the construction of rich models of a discourse, where all the information are posited in relation with the others in a way to preserve coherence. Cognitive conflicts and socio-cognitive conflicts produce lack of coherence in our representations and therefore induce us to modify our mental representations. The general lesson that we might derive from the argumentation above is that multiple perspectives on a single task may prove useful for deep learning.

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References

- Alibali MW, Goldin-Meadow S (1993) Gesture-speech mismatch and mechanisms of learning: what the hands reveal about a child's state of mind. *Cogn Psychol* 25:468–523
- Alibali MW, Kita S, Bigelow LJ, Wolfman CM, Klein SM (2001) Gesture plays a role in thinking for speaking, vol 18. *Actes du colloque*, Orange, Aix-en-Provence, pp 407–410
- Alibali MW, Kita S, Young AJ (2000) Gesture and the process of speech production: we think, therefore we gesture. *Lang Cogn Proc* 15:593–613
- Anderson JR (1983) *The architecture of cognition*. Harvard University Press, Cambridge
- Azmitia M (1988) Peer interaction and problem solving: when are two heads better than one? *Child Dev* 59:87–96

- Antinucci F, Troise A, Natale F, Berlinguer L (1994) How does hypermedia affect learning? In: Levonen J, Tukiainen M (eds) *Complex learning in computer environments*. TOTY, Joensuu, pp 74–77
- Bara BG (1995) *Cognitive science: a developmental approach to the simulation of the mind*. Lawrence Erlbaum Associates, Hove
- Bara B, Bucciarelli M, Geminiani G (2000) Development and decay of extra-linguistic communication. *Brain Cogn* 43:1–3
- Baron-Cohen S (1994) How to build a baby that can read minds: cognitive mechanism in mind reading. *Cah Psychol Cogn* 13:1–40
- Biggs JB (2001) Enhancing learning: a matter of style or approach? In: Sterneberg RJ, Zhang LF (eds) *Perspectives on thinking, learning, and cognitive styles*. Lawrence Erlbaum Associates, Mahwah, pp 73–102
- Bruner JS, Olver RR, Greenfield PM (1966) *Studies on cognitive growth*. Wiley, New York
- Bucciarelli M (2005) Pointing and communicative meanings. *Curr Anthropol* 46(1):152–153
- Bucciarelli M, Colle L, Bara BG (2003) How children comprehend speech acts and communicative gestures. *J Pragmatics* 35:207–241
- Bucciarelli M, Johnson-Laird PN (2001) Falsification and the role of the theory of mind in the reduced array selection task. *Curr Psychol Lett* 4:7–22
- Byrne RMJ, Handley SJ (1997) Reasoning strategies for suppositional deductions. *Cognition* 62:1–49
- Caravita S, Halldén O (1994) Re-framing the problem of conceptual change. *Learn Instrum* 4:89–111
- Carney RN, Levin JR (2002) Pictorial illustrations still improve students' learning from text. *Educ Psychol Rev* 14(1):5–26
- Chan C (2001) Peer collaboration and discourse patterns in learning from incompatible information. *Instrum Sci* 29:443–479
- Cheng PW, Holyoak KJ (1989) On the natural selection of reasoning theories. *Cognition* 33:285–314
- Church RB (1999) Using gesture and speech to capture transitions in learning. *Cogn Dev* 14:313–342
- Clement JJ, Steinberg MS (2002) Step-wise evolution of mental models of electric circuits: a “learning-aloud” case study. *J Learn Sci* 11:389–452
- Craik FIM, Tulving E (1975) Depth of processing and the retention of words in episodic memory. *J Exp Psychol Gen* 104:268–294
- Cuevas HM, Fiore SM, Oser RL (2002) Scaffolding cognitive and metacognitive processes in low verbal ability learners: use of diagrams in computer-based training environments. *Instrum Sci* 30(6):433–464
- Cummins DD (1996) Evidence of deontic reasoning in 3- and 4-year-old children. *Mem Cogn* 24(6):823–829
- Cutica I, Bucciarelli M (2003) Gestures and the construction of models. In: *Proceedings of the second international conference on reasoning and decision making. Reasoning and understanding: mental models, relevance and limited rationality approaches*, March 17–18, Padova, p 11
- Cutica I, Bucciarelli M (2007) Gestures and mental models in learning from discourse (Submitted)
- Doise W, Mugny G (1984) *The social development of the intellect*. Pergamon, Oxford
- Druyan S (2001) A comparison of four types of cognitive conflict and their effect on cognitive development. *Int J Behav Dev* 25:226–236
- Fuchs LS, Fuchs D, Kazdan S, Karns K, Calhoon MB, Hamlett CL, Hewlett S (2000) Effects of workgroup structure and size on student productivity during collaborative work on complex tasks. *Elem School J* 100:183–212
- Garber P, Alibali MW, Goldin-Meadow S (1998) Knowledge conveyed in gesture is not tied to the hands. *Child Dev* 69:75–84
- Garber P, Goldin-Meadow S (2002) Gesture offers insight into problem-solving in children and adults. *Cogn Sci* 26:817–831
- García-Madruga J, Carriedo N, González-Labra MJ (eds) (2000) *Mental models in reasoning*. UNED, Madrid
- Giroto V, Light P, Colbourn CJ (1988) Pragmatic schemas and conditional reasoning in children. *Q J Exp Psychol* 40A:469–482
- Glenberg AM, Langston WE (1992) Comprehension of illustrated text: pictures help to build mental models. *J Mem Lang* 31:129–151

- Goldin-Meadow S (2001) Giving the mind a hand: the role of gesture in cognitive change. In: McClelland J, Siegler RS (eds) *Mechanisms of cognitive development. Behavioral and neural perspectives*. Lawrence Erlbaum Associates, Mahwah, pp 5–31
- Goldin-Meadow S, Butcher C (2003) Pointing toward two-word speech in young children. In: Kita S (ed) *Pointing. Where language, culture, and cognition meet*. Lawrence Erlbaum Associates, Mahwah, pp 85–107
- Goldin-Meadow S, Sandhofer CM (1999) Gestures convey substantive information about a child's thoughts to ordinary listeners. *Dev Sci* 2:67–74
- Griggs RA, Cox JR (1982) The elusive thematic-materials effect in Wason's selection task. *Br J Psychol* 73:407–420
- Gyselinck V (1996) Illustrations and mental models in text comprehension. *Ann Psychol* 96:495–516
- Hadamard J (1945) *The psychology of invention in the mathematical field*. Dover, New York
- Halford G (1993) *Children's understanding: the development of mental models*. Lawrence Erlbaum Associates, Hillsdale
- Johnson-Laird PN (1983) *Mental models: towards a cognitive science of language, and consciousness*. Cambridge University Press, Cambridge
- Johnson-Laird PN (2001) Mental models and human reasoning. In: Dupoux E (ed) *Language, brain, and cognitive development: essays in honor of Jacques Mehler*. The MIT Press, Cambridge, pp 85–102
- Johnson-Laird PN, Bethell-Fax C (1978) Memory for questions and amount of processing. *Mem Cogn* 6:496–501
- Johnson-Laird PN, Byrne RMJ (1991) *Deduction*. Lawrence Erlbaum Associates, London
- Johnson-Laird PN, Byrne RMJ (2002) Conditionals: a theory of meaning, pragmatics, and inference. *Psychol Rev* 109:646–678
- Johnson-Laird PN, Legrenzi P, Sonino-Legrenzi M (1972) Reasoning and a sense of reality. *Br J Psychol* 63:395–400
- Johnson-Laird PN, Savary F, Bucciarelli M (2000) Strategies and tactics in reasoning. In: Schaeken WS, Vandierendonck A, De Vooght G, d'Ydewalle G (eds) *Deductive reasoning and strategies*. Lawrence Erlbaum Associates, Mahwah, pp 209–240
- Karmiloff-Smith A (1992) *Beyond modularity: a developmental perspective on cognitive science*. The MIT Press, Cambridge
- Kelly GA (1955) *The psychology of personal construct*. Norton, New York
- Kelly S, Church RB (1998) A comparison between children's and adults' ability to detect conceptual information conveyed through representational gestures. *Child Dev* 69:85–93
- Krauss RM, Morrel-Samuels P, Colasante C (1991) Do conversational hand gestures communicate? *J Pers Soc Psychol* 61:743–754
- Kroger K, Cheng PW, Holyoak KJ (1993) Evoking the permission schema: the impact of explicit negation and a violation-checking context. *Q J Exp Psychol* 46A:615–635
- Kuhn D (1989) Children and adults as intuitive scientists. *Psychol Rev* 96:674–689
- Kuhn D (2000) Metacognitive development. *Curr Dir Psychol Sci* 9(5):178–181
- Kuhn D, Kats JB, Dean D (2004). Developing reason. *Think Reason* 10:197–219
- Legrenzi P, Butera F, Mugny G, Pérez JA (1991) Majority and minority influence in inductive reasoning. *Eur J Soc Psychol* 21:359–363
- Leslie AM (1987) Pretense and representation: the origins of "theory of mind". *Psychol Rev* 94:412–426
- Light P, Glachan M (1985) Facilitation of individual problem solving through peer interaction. *Educ Psychol* 5:217–225
- Limón M (2001) On the cognitive conflict as an instructional strategy for conceptual change: a critical appraisal. *Learn Instrum* 11:357–380
- Limón M, Carretero M (1997) Conceptual change and anomalous data: a case study in the domain of natural sciences. *Eur J Psychol Educ* 12(2):213–230
- Manktelow KI, Over DE (1990) *Inference and understanding*. Frances-Routledge, Florence
- Manktelow KI, Over DE (1991) Social roles and utilities in reasoning with deontic conditionals. *Cognition* 39:85–105
- Mayer RE (1997) Multimedia learning: are we asking the right questions? *Educ Psychol* 32:1–19
- Mayer RE, Moreno R (2002) Aids to computer-based multimedia learning. *Learn Instrum* 12:107–119

- McClelland JL (1995) A connectionist perspective on knowledge and development. In: Simon TJ, Halford GS (eds) *Developing cognitive competence. New approaches to process modeling*. Lawrence Erlbaum Associates, Hove, pp 157–204
- McKendree J, Small C, Stenning K, Conlon T (2002) The role of representation in teaching and learning critical thinking. *Educ Rev* 54:57–67
- Montgomery H (1984) Mental models and problem solving: three challenges to a theory of restructuring and insight. *Scand J Psychol* 29:85–94
- Moreno R, Mayer RE (1999) Cognitive principles of multimedia learning: the role of modality and contiguity. *J Educ Psychol* 91:358–368
- Moreno R, Mayer RE (2002) Verbal redundancy in multimedia learning: when reading helps listening. *J Educ Psychol* 94:156–163
- Perret-Clermont AN (1980) Social interaction and cognitive development in children. Academic, London
- Popper K (1982) In: Bartely WW III (ed) *Postscript to “the logic of scientific discovery”*. Hutchinson, London
- Premack D, Woodruff G (1978) Does the chimpanzee have a theory of mind? *Behav Brain Sci* 1:515–526
- Rickheit G, Wrobel H (1999) Cognitive aspects of coordination processes. In: Rickheit G, Habel C (eds) *Mental models in discourse processing and reasoning*. North-Holland/Elsevier, Amsterdam, pp 113–130
- Roazzi A, Bryant P (1998) The effects of symmetrical and asymmetrical social interaction on children’s logical inferences. *Br J Dev Psychol* 16:175–181
- Russell J (1982) Cognitive conflict transmission and justification: conservation attainment through dyadic interactions. *J Genet Psychol* 140:283–297
- Sacco K, Bucciarelli M (2007) The role of cognitive and socio-cognitive conflict in learning to reason. *Mind Soc* (in press)
- Schnotz W, Bannert M (1999) Influence of the type of visualization on the construction of mental models during picture and text comprehension. *Z Exp Psychol* 46:217–236
- Schwartz DL, Black JB (1996) Shuttling between depictive models and abstract rules: induction and fallback. *Cogn Sci* 20:457–497
- Siegler RS (1976) Three aspects of cognitive development. *Cogn Psychol* 8:481–520
- Siegler RS (1995) How does change occur: a microgenetic study of number conservation. *Cogn Psychol* 28:225–273
- Simon TJ, Halford SG (1995) Computational models and cognitive change. In: Simon TJ, Halford SG (eds) *Developing cognitive competence. New approaches to process modeling*. Lawrence Erlbaum Associates, Hove, pp 1–30
- Spiro RJ, Jehng JC (1990) Cognitive flexibility & hypertext: theory & technology for the nonlinear & multidimensional traversal of complex subject matter. In: D Nix R Spiro (eds) *Cognition, education, & multimedia: exploring ideas in high technology*. Lawrence Erlbaum Associates, Hillsdale, pp 163–205
- Stanovich KE, West RF (1998) Cognitive ability and variation in selection task performance. *Think Reason* 4:193–230
- Stenning K (2000) Distinctions with differences: comparing criteria for distinguishing diagrammatic from sentential systems. Theory and application of diagrams. *Lect Notes Artif Int* 1899:132–148
- Vandenplas-Holper C (1996) Intraindividual and interindividual cognitive conflict, related variables and relations with cognitive development. *Swiss J Psychol* 55(2/3):161–175
- Vekiri I (2002) What is the value of graphical displays in learning? *Educ Psychol Rev* 14(3):261–312
- Vosniadou S (ed) (1994) Conceptual change in the physical sciences. *Learn Instrum* 4 (special issue)
- Vygotsky LS (1978) *Mind in society: the development of higher psychological processes*. Harvard University Press, Cambridge
- van Dijk IA, Kintsch W (1983) *Strategies of discourse comprehension*. Academic, New York
- Wason P (1966) Reasoning. In: Foss BM (ed) *New horizons in psychology*. Penguin, Harmondsworth, pp 135–151
- Wason P (1968) Reasoning about a rule. *Q J Exp Psychol* 20:273–281